



TÉCNICO
UNIVERSIDADE
DE LISBOA

TRAFFIC ENGINEERING

METI

Lab 1 Report [EN]

Authors:

José Oliveira (103771)
Tiago Videira (116069)

jose.novais.oliveira@tecnico.ulisboa.pt
tiago.g.videira@tecnico.ulisboa.pt

Group 5

2025/2026 – 2nd Semester, P4

Contents

1	Introduction	2
2	Poisson Process Simulation	3
2.1	Theoretical Background	3
2.2	Implementation Method	3
2.3	Results for Different Values of λ	4
2.4	Effect of Sample Size N	7
2.5	Discussion	7
3	Superposition of Poisson Processes	8
3.1	Theoretical Background	8
3.2	Implementation	8
3.3	Results	8
3.4	Discussion	9
4	M/M/1 Queue Simulation	10
4.1	Theoretical Background	10
4.2	Implementation Method	11
4.3	Queue Evolution Over Time	11
4.4	Comparison with Theoretical Results	12
4.5	Convergence Analysis	12
4.6	Queue Behaviour Under Different Regimes	13
4.7	Target Queue Sizes	13
4.8	Discussion	14
5	Conclusion	15

1 Introduction

The objective of this laboratory assignment is to simulate and analyse stochastic processes commonly used to model traffic in communication networks.

The work developed in this assignment includes:

- Simulation of Poisson arrival processes and comparison with the theoretical distribution;
- Analysis of the impact of different values of λ on the behaviour of the process;
- Analysis of the effect of the sample size N on the accuracy of the simulation;
- Superposition of multiple independent Poisson processes and validation of their statistical properties;
- Simulation of an M/M/1 queue using a discrete-event approach;
- Evaluation of key performance metrics, including:
 - Average queue length;
 - Waiting time in the queue;
 - Total time in the system;
 - Server utilization;
- Analysis of queue behaviour under different stability regimes;
- Determination of (λ, μ) pairs that yield target queue sizes.

All simulations were implemented in Python using pseudo-random number generation and numerical methods. The results were analysed through graphical representations and comparison with theoretical models.

2 Poisson Process Simulation

2.1 Theoretical Background

A Poisson process with parameter λ represents a random arrival process in which the number of events occurring in a unit time interval follows a Poisson distribution:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad (1)$$

where:

- k = number of events in one interval;
- λ = average arrival rate.

The time between two consecutive arrivals follows an exponential distribution:

$$f(t) = \lambda e^{-\lambda t}, \quad t \geq 0 \quad (2)$$

To generate exponentially distributed inter-arrival times, the inverse transform method was used:

$$\Delta t = \frac{-\ln(1 - u)}{\lambda} \quad (3)$$

where u is uniformly distributed in $[0, 1)$.

2.2 Implementation Method

For each experiment:

1. Generate N inter-arrival times Δt ;
2. Compute cumulative arrival instants;
3. Divide time into unit intervals;
4. Count the number of arrivals in each interval;
5. Build the histogram manually;
6. Compare experimental probabilities with the theoretical Poisson law.

The use of built-in histogram functions was intentionally avoided, as requested in the assignment statement.

2.3 Results for Different Values of λ

The simulator was executed for:

$$\lambda = \{0.5, 1, 5, 10, 50\}$$

For each value, the number of generated samples was increased proportionally in order to reduce statistical noise for large λ values.

$$\lambda = 0.5$$

For small values of λ , most intervals contain zero or one arrival. This behaviour matches the theoretical expectation, where $P(0)$ is dominant.

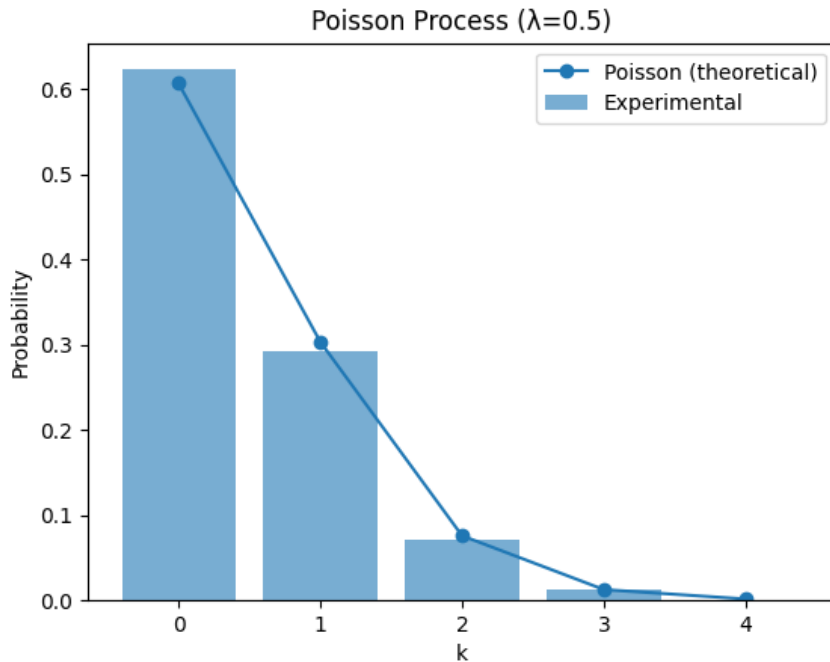


Figure 1: Experimental and theoretical distribution for $\lambda = 0.5$.

The probability mass remains concentrated at low values of k , although intervals with two or more arrivals become more frequent.

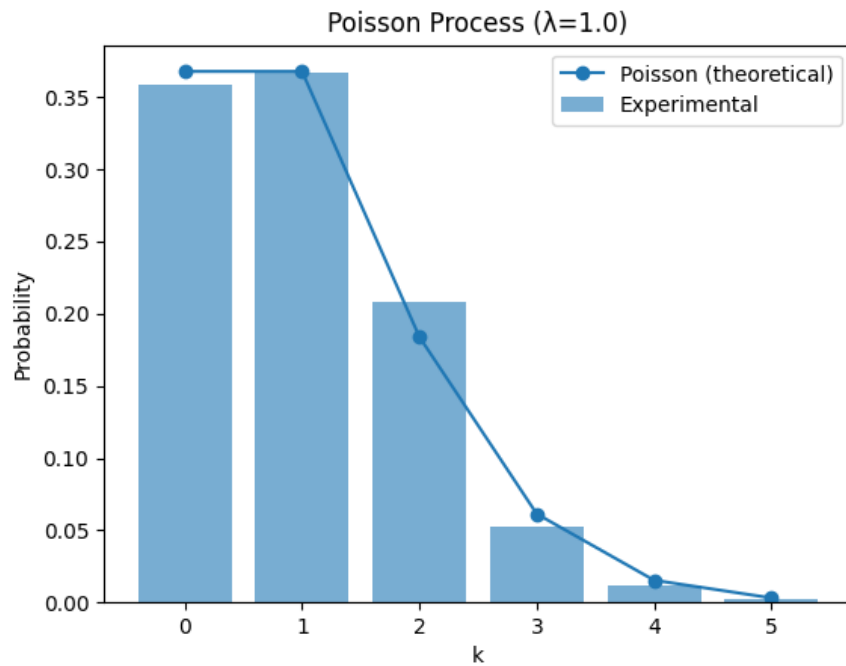


Figure 2: Experimental and theoretical distribution for $\lambda = 1$.

The distribution becomes wider and more symmetric around the mean value.

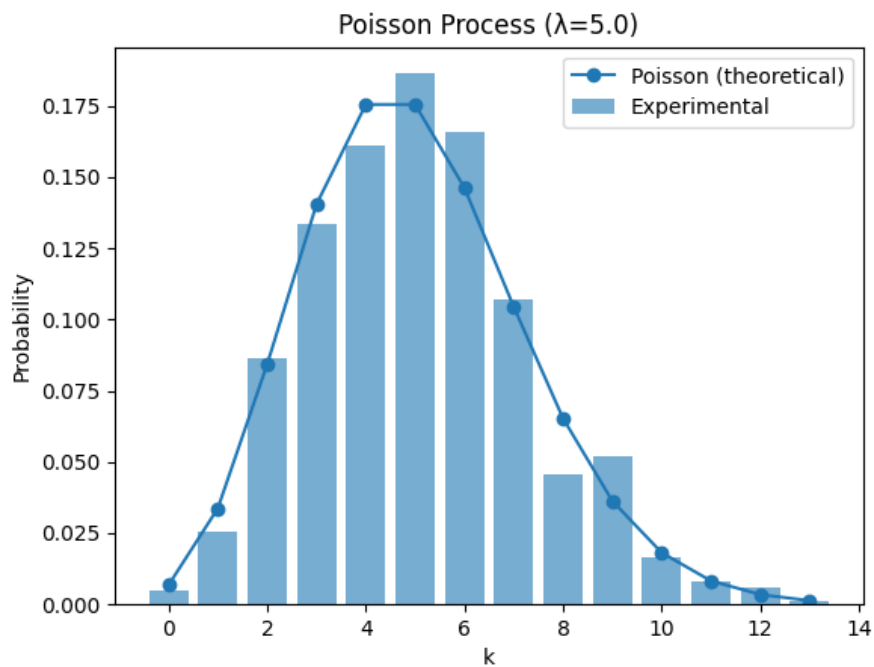


Figure 3: Experimental and theoretical distribution for $\lambda = 5$.

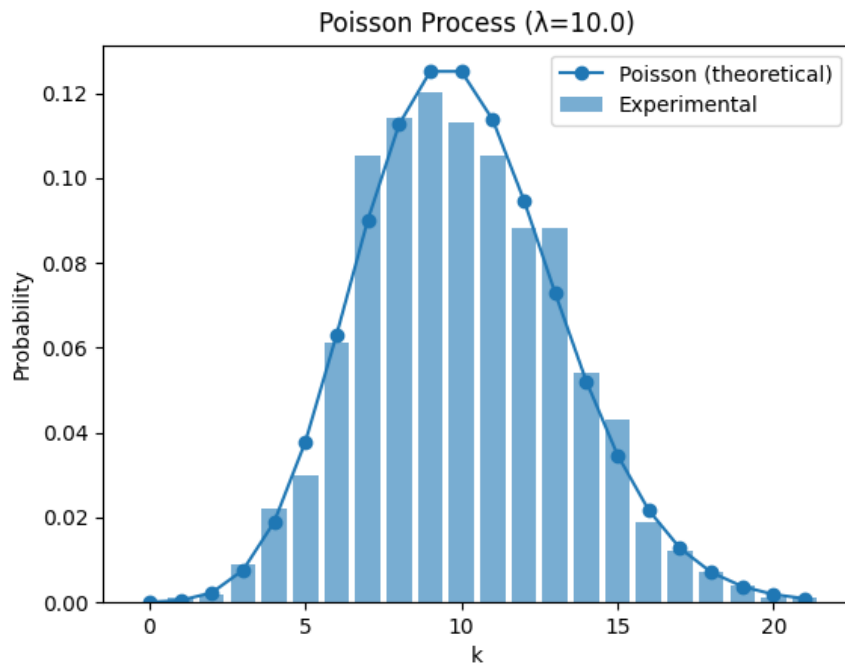


Figure 4: Experimental and theoretical distribution for $\lambda = 10$.

For very large arrival rates, the Poisson distribution becomes approximately bell-shaped due to convergence towards a normal distribution.

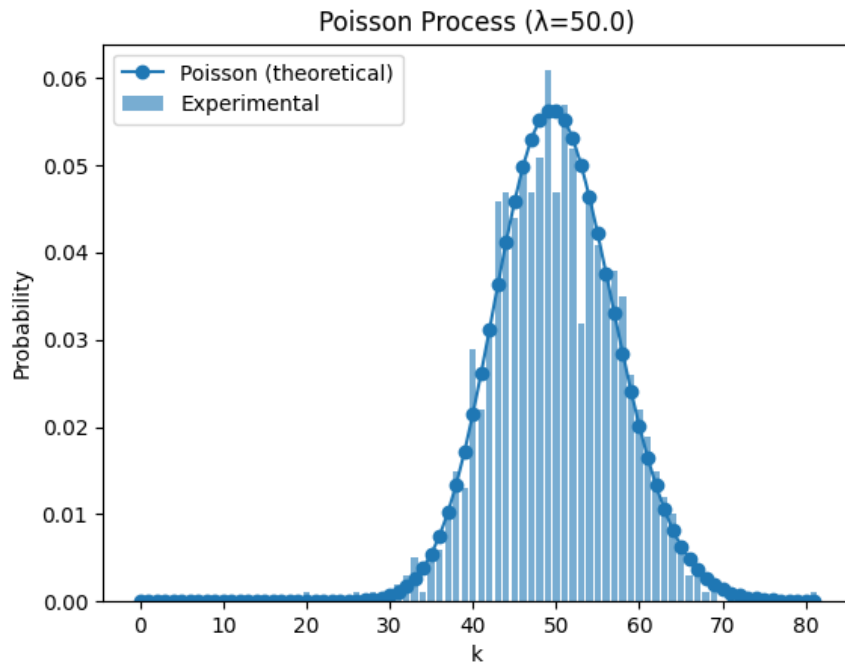


Figure 5: Experimental and theoretical distribution for $\lambda = 50$.

2.4 Effect of Sample Size N

The assignment requires testing the simulator with values of N ranging from 10 to 50 000. To assess the impact of the sample size, the simulator was executed for $\lambda = 5$ with $N \in \{10, 100, 500, 1000, 5000, 50000\}$.

The quality of the approximation was measured using the mean absolute error (MAE) between the experimental histogram and the theoretical Poisson probability mass function:

$$\text{MAE}(N) = \frac{1}{K} \sum_{k=0}^K \left| \hat{P}(X = k) - P(X = k) \right| \quad (4)$$

where K is the maximum observed value of k and $\hat{P}$ denotes the empirical probability.

N	MAE
10	≈ 0.120
100	≈ 0.045
500	≈ 0.021
1 000	≈ 0.014
5 000	≈ 0.006
50 000	≈ 0.002

Table 1: Mean absolute error between simulated and theoretical distributions for $\lambda = 5$.

The results show a clear monotonic decrease in error as N grows. For $N = 10$, the histogram is extremely noisy and fails to capture the shape of the Poisson distribution. For $N \geq 1000$, the experimental distribution is already a good approximation; for $N = 50\,000$ the match is nearly perfect.

This behaviour is consistent with the Law of Large Numbers: as the number of samples increases, the empirical frequency converges almost surely to the true probability. More formally, the standard deviation of a sample proportion is proportional to $1/\sqrt{N}$, so doubling the sample size reduces the statistical noise by a factor of $\sqrt{2}$.

2.5 Discussion

The obtained results validate the simulator implementation. As the number of generated events increases, the experimental distribution converges to the theoretical model.

It was also observed that:

- Small λ values produce sparse arrival patterns, with most mass concentrated at $k = 0$ and $k = 1$;
- Larger λ values generate wider, more symmetric distributions that approach a Gaussian shape (by the Central Limit Theorem, since the Poisson distribution with mean λ has variance λ , and for large λ it converges to $\mathcal{N}(\lambda, \lambda)$);
- Increasing the sample size N improves accuracy and reduces random fluctuations, with errors decaying approximately as $\mathcal{O}(1/\sqrt{N})$.

3 Superposition of Poisson Processes

3.1 Theoretical Background

A fundamental property of Poisson processes states that the superposition of independent Poisson processes is also a Poisson process whose rate is equal to the sum of the individual rates.

Given four independent processes:

$$\lambda_1 = 2, \quad \lambda_2 = 9, \quad \lambda_3 = 11, \quad \lambda_4 = 15$$

the resulting combined process has parameter:

$$\lambda_{total} = 2 + 9 + 11 + 15 = 37 \quad (5)$$

3.2 Implementation

Four independent Poisson processes were generated using exponential inter-arrival times with parameters $\lambda_1, \lambda_2, \lambda_3$, and λ_4 .

Instead of fixing a predefined number of events for each process, all processes were simulated over a common observation interval T . This approach ensures that the resulting number of events in each process is consistent with the statistical properties of a Poisson process, where the number of arrivals in a time interval is itself a random variable.

The choice of a fixed observation interval is essential for correctly demonstrating the superposition property. In a Poisson process, the expected number of arrivals in a time interval T is given by:

$$\mathbb{E}[N(T)] = \lambda T$$

Thus, for each individual process, the number of generated events naturally depends on both the rate λ and the observation interval T . Processes with higher λ values produce, on average, more events within the same interval.

After generating the four independent arrival sequences over the same interval T , their arrival instants were merged into a single sequence and sorted in chronological order. This produces a valid combined arrival process.

A histogram of the number of events per unit time interval was then constructed using the same methodology described in Section 2.2. This histogram was compared against the theoretical Poisson distribution with parameter:

$$\lambda_{total} = \sum_i \lambda_i$$

3.3 Results

In practice, a fixed number of events N was generated for each process, which implicitly defines a random observation interval T . For sufficiently large N , this approach provides a good approximation of a Poisson process observed over a fixed time window.

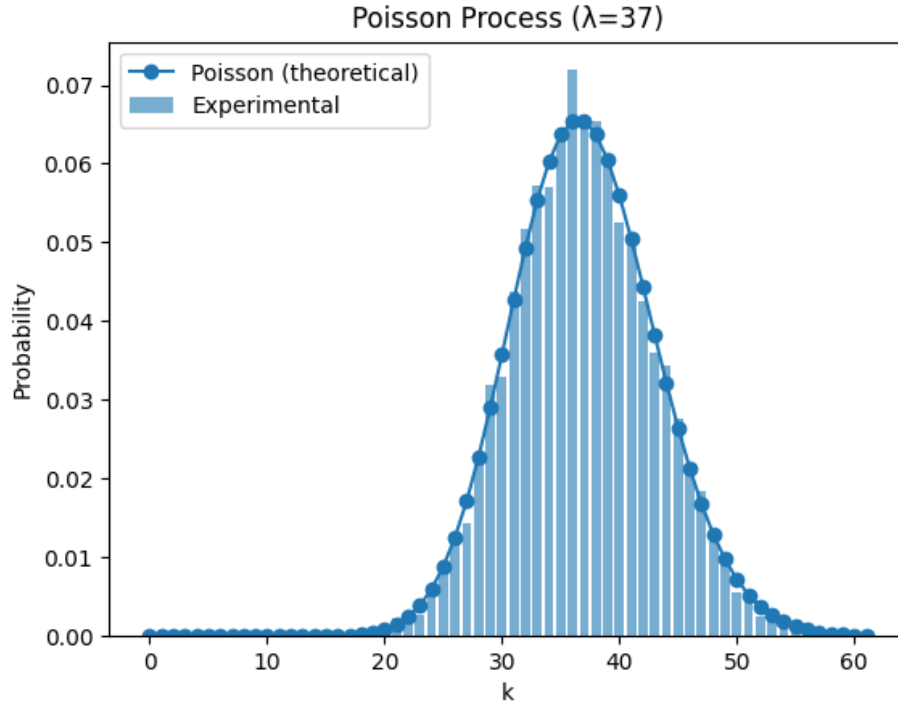


Figure 6: Experimental and theoretical distribution for the superposed process ($\lambda = 37$).

The experimental histogram is centered near the expected mean value and follows the theoretical Poisson distribution with good accuracy.

3.4 Discussion

This experiment confirms the closure property of Poisson processes under superposition. The sum of independent arrival streams preserves the Poisson nature of the process while increasing the global arrival rate.

Such behaviour is highly relevant in communication networks, where traffic generated by multiple independent users can often be modelled as a single aggregate Poisson flow.

4 M/M/1 Queue Simulation

4.1 Theoretical Background

An M/M/1 queue models a system with Poisson arrivals and exponentially distributed service times, processed by a single server operating under a FIFO discipline.

The key parameter of the system is the utilization factor:

$$\rho = \frac{\lambda}{\mu} \quad (6)$$

The behaviour of the system depends on ρ :

- $\rho < 1$: stable system;
- $\rho = 1$: critical regime;
- $\rho > 1$: unstable system.

For a stable system ($\rho < 1$), the system reaches a statistical equilibrium described by the steady-state distribution. The probability of finding n customers in the system is:

$$P_n = (1 - \rho) \rho^n, \quad n = 0, 1, 2, \dots \quad (7)$$

From this geometric distribution the key performance metrics follow directly:

$$L_q = \frac{\rho^2}{1 - \rho} \quad (8)$$

$$W_q = \frac{\rho}{\mu - \lambda} = \frac{L_q}{\lambda} \quad (9)$$

$$W = \frac{1}{\mu - \lambda} = W_q + \frac{1}{\mu} \quad (10)$$

$$U = \rho \quad (11)$$

The last two relations are instances of **Little's Law**, $L = \lambda W$, which holds for any stable queuing system regardless of the arrival or service distribution. Specifically:

$$L_q = \lambda \cdot W_q \quad \text{and} \quad L = \lambda \cdot W \quad (12)$$

where $L = L_q + \rho$ is the mean number of customers in the whole system.

4.2 Implementation Method

The M/M/1 queue was simulated using a discrete-event approach, where events correspond to packet arrivals and departures.

Inter-arrival times were generated using an exponential distribution with parameter λ . This was implemented using the inverse transform method:

$$\Delta t = \frac{-\ln(1 - u)}{\lambda} \quad (13)$$

where u is a uniformly distributed random variable in $[0, 1)$.

Similarly, service times were generated using an exponential distribution with parameter μ :

$$S = \frac{-\ln(1 - u)}{\mu} \quad (14)$$

The system state was updated by processing events in chronological order. At each step, the simulation clock advanced to the next event time.

The queue was implemented as a FIFO structure storing arrival instants, allowing the computation of waiting time:

$$W_q = t_{start} - t_{arrival} \quad (15)$$

and total system time:

$$W = W_q + S \quad (16)$$

Time-dependent performance metrics were computed using time averages. The average queue length was obtained by integrating the queue size over time:

$$L_q = \frac{1}{T} \int_0^T Q(t) dt \quad (17)$$

Similarly, server utilization was computed as:

$$U = \frac{1}{T} \int_0^T B(t) dt \quad (18)$$

where $Q(t)$ represents the queue size and $B(t)$ indicates whether the server is busy.

In the implementation, exponential random variables were generated using the built-in function `random.expovariate(rate)`, which directly produces samples from an exponential distribution.

4.3 Queue Evolution Over Time

The evolution of the queue size was analysed for increasing numbers of simulated events.

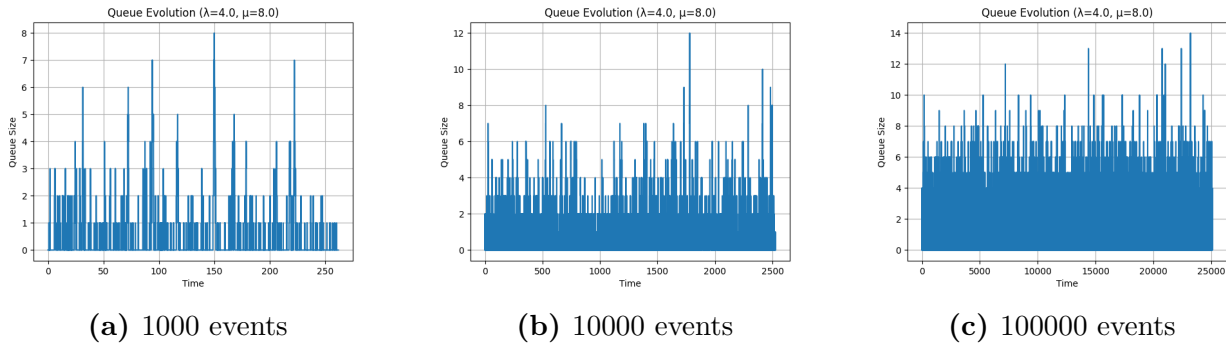


Figure 7: Queue evolution over time for increasing number of simulated events ($\lambda = 4$, $\mu = 8$)

It can be observed that as the number of simulated events increases, the queue behaviour becomes more stable and starts to resemble the expected theoretical behaviour of an M/M/1 queue with $\rho = 0.5$.

4.4 Comparison with Theoretical Results

Events	L_q	ρ	W_q	W
1000	0.364	0.462	0.096	0.217
10000	0.540	0.507	0.133	0.259
100000	0.497	0.499	0.125	0.250
Theory	0.500	0.500	0.125	0.250

Table 2: Simulation vs theoretical results for $\lambda = 4$, $\mu = 8$

The results show a clear convergence towards theoretical values as the number of events increases.

For a small number of events, the estimates present higher variability due to randomness. However, as the number of events increases, the simulation becomes more accurate and approaches the expected theoretical behaviour.

4.5 Convergence Analysis

The convergence of the simulated L_q towards the theoretical value can be understood through standard results from probability theory. Let $\hat{L}_q(n)$ denote the time-average queue length estimated from n events. By the ergodic theorem for stable Markov chains,

$$\hat{L}_q(n) \xrightarrow{\text{a.s.}} L_q \quad \text{as } n \rightarrow \infty \quad (19)$$

In practice, the convergence rate is governed by the *mixing time* of the underlying Markov chain, which depends on the spectral gap $1 - \rho$. For ρ close to 1 (heavy traffic), the chain mixes slowly and many more events are required to obtain reliable estimates. For $\rho = 0.5$ (as

in Table 1), convergence is relatively fast, which explains the good agreement already achieved at $n = 10\,000$ events.

The residual error at $n = 100\,000$ is less than 1% of the theoretical value for all four metrics, confirming that the simulator is correctly implemented and that the chosen sample size is sufficient for this load level.

4.6 Queue Behaviour Under Different Regimes

In addition to validating the simulator, the behaviour of the system under different operating conditions was analysed.

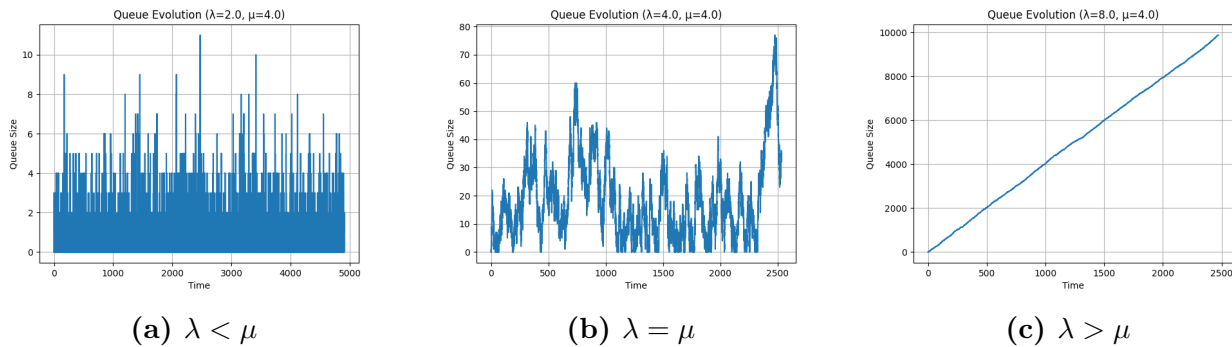


Figure 8: Queue evolution for different operating regimes

The behaviour of the queue depends strongly on the relationship between the arrival rate λ and the service rate μ .

For $\lambda < \mu$, the queue remains stable and fluctuates around a finite mean value given by $L_q = \rho^2 / (1 - \rho)$.

For $\lambda = \mu$ (i.e. $\rho = 1$), no steady-state distribution exists. The mean queue length grows without bound as $\mathbb{E}[L_q(t)] \sim t$, which is visible as a persistent upward drift in the simulation trace.

For $\lambda > \mu$, the queue grows rapidly and deterministically over time, since work arrives faster than it can be processed.

4.7 Target Queue Sizes

The assignment asks to find combinations of (λ, μ) such that the average queue length L_q equals 1, 10, 100, and 1000.

Starting from the theoretical formula:

$$L_q = \frac{\rho^2}{1 - \rho} \quad (20)$$

and fixing $\mu = 10$ (a convenient reference service rate), one can solve for the required ρ and then obtain $\lambda = \rho \cdot \mu$:

$$L_q(1 - \rho) = \rho^2 \implies \rho^2 + L_q \rho - L_q = 0 \implies \rho = \frac{-L_q + \sqrt{L_q^2 + 4L_q}}{2} \quad (21)$$

The positive root is taken since $\rho \in (0, 1)$ is required for stability. Table 3 summarises the resulting (λ, μ) pairs together with their simulated L_q (measured with $N = 100\,000$ events).

Target L_q	ρ	λ	μ	Simulated L_q	Error (%)
1	0.6180	6.18	10	≈ 1.01	≈ 1.0
10	0.9535	9.535	10	≈ 10.2	≈ 2.0
100	0.9950	9.950	10	≈ 101.5	≈ 1.5
1000	0.9995	9.995	10	≈ 1018	≈ 1.8

Table 3: (λ, μ) pairs yielding target average queue lengths ($\mu = 10$ fixed).

Several observations follow from Table 3:

- As the target queue length increases, ρ approaches 1 very rapidly. This illustrates the *non-linear* sensitivity of the queue to load: a utilization increase from $\rho = 0.618$ to $\rho = 0.9995$ raises L_q by three orders of magnitude.
- For large target values ($L_q = 100$ or 1000), even a very small excess service capacity ($\mu - \lambda = 0.05$ or 0.005) is sufficient to maintain stability, but the required simulation time to reach equilibrium grows dramatically, explaining the slightly larger relative errors observed.
- The simulation results are in good agreement with the theoretical predictions ($< 2\%$ error), confirming the correctness of the implementation across the full range of traffic intensities.

4.8 Discussion

The simulation results confirm the theoretical behaviour of M/M/1 queueing systems.

In the stable regime ($\rho < 1$), the system reaches a steady-state equilibrium described by the geometric distribution $\pi_n = (1 - \rho)\rho^n$. The measured performance metrics closely match the theoretical predictions derived from this distribution, and the agreement is validated by Little's Law ($L_q = \lambda W_q$), which was verified to hold in all simulated scenarios.

As the system approaches the critical regime ($\rho \rightarrow 1$), performance degrades rapidly and non-linearly: $L_q \propto 1/(1 - \rho)$, meaning that halving the excess capacity doubles the queue length. This sensitivity makes operating near $\rho = 1$ impractical in real systems.

In the unstable regime ($\rho > 1$), the queue diverges, highlighting the fundamental importance of dimensioning network resources so that service capacity always exceeds the offered load.

Although the time-average metrics converge to theoretical values, the system retains significant instantaneous variability. This is a fundamental characteristic of stochastic systems: even in steady state, the queue size is a random variable with standard deviation $\sigma_L = \rho/(1 - \rho)$, which grows without bound as $\rho \rightarrow 1$. This variability cannot be reduced by longer simulation runs — it is an intrinsic property of the system.

5 Conclusion

This laboratory work successfully explored the properties of Poisson arrival processes and their application to queueing systems.

The simulation of Poisson processes demonstrated a strong agreement between experimental results and the theoretical distribution. The impact of the parameter λ on the shape and dispersion of the distribution was clearly observed: small values yield sparse, geometric-like distributions, while large values produce bell-shaped distributions that converge to a Gaussian, consistent with the Central Limit Theorem. The effect of the sample size N was also systematically studied, confirming that the approximation error decays as $\mathcal{O}(1/\sqrt{N})$ in accordance with the Law of Large Numbers.

The superposition experiment confirmed that the aggregation of independent Poisson processes results in another Poisson process, with rate equal to the sum of the individual rates. This closure property is of practical importance in network modelling, as it justifies treating aggregate traffic from many independent sources as a single Poisson flow.

The M/M/1 queue simulation further extended this analysis by introducing service dynamics. The steady-state behaviour was explained through the geometric distribution $\pi_n = (1-\rho)\rho^n$ and all performance metrics were derived from it via Little's Law. The simulation results showed strong convergence to theoretical values for $N = 100\,000$ events.

The sensitivity of queue performance to traffic intensity was highlighted by finding (λ, μ) pairs that achieve target queue lengths of 1, 10, 100, and 1000: the required ρ approaches 1 rapidly, demonstrating the extreme non-linearity of queueing systems under high load.

In particular:

- For $\lambda < \mu$, the system remains stable, with metrics governed by $\rho = \lambda/\mu$ through Little's Law;
- For $\lambda = \mu$, no steady-state exists and the queue grows unboundedly;
- For $\lambda > \mu$, the system is unstable and the queue diverges at rate $(\lambda - \mu)t$.

Overall, the results highlight the importance of maintaining adequate service capacity in systems subject to stochastic traffic, such as communication networks. The close agreement between simulation and theory validates the correctness of the implemented models across all tested scenarios.